



IR-HyChem: Towards modeling the high-T combustion behavior of aviation fuels using infrared spectra

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ABSTRACT

A Fourier transform infrared (FTIR) spectra-based approach, namely IR-HyChem, was developed to model the combustion behavior of jet and rocket fuels. Earlier shock-tube experiments employed laser absorption spectroscopy (LAS) to measure the yields of key stable intermediates: CH₄, C₂H₄, and >C₂ alkenes such as C₃H₆, 1-C₄H₈ and *i*-C₄H₈, during the pyrolysis of neat hydrocarbons across several molecular classes (*n*-alkanes, lightly branched alkanes and highly branched alkanes). These measurements revealed empirical relations of molecular structure to the yields of these intermediates. The relationships provided important insights into fuel reactivity under high-temperature, combustor-relevant conditions. The IR-HyChem methodology establishes quantitative correlations between spectral features and the yields of these pyrolysis intermediates. Using this framework, IR-HyChem models were demonstrated for two jet fuels (JP-8 and F-24) and a rocket fuel (RP-1), by constraining a subset of stoichiometric parameters in the HyChem lumped reactions, resulting in partially constrained IR-HyChem models. These models were evaluated against ignition delay times (IDTs) measured behind reflected shock waves at elevated pressures, demonstrating strong agreement with experimental data. A Monte Carlo uncertainty analysis revealed that imposing FTIR-based constraints reduced variance in IDT predictions compared to unconstrained models. Furthermore, sensitivity analysis indicated that additional IR spectra-based correlations could improve the accuracy of the IR-HyChem models. Overall, this work demonstrates the utility of FTIR spectra and their potential use as a low-volume tool for developing predictive chemistry models for the combustion of real, multi-component fuels.

1. Introduction

The transportation sector is one of the largest contributors to global greenhouse gas emissions. Although aviation-related emissions only make up about 11 % of the overall transportation sector's share, studies analyzing net-zero emissions energy systems indicate that emissions from aviation are particularly difficult to eliminate [1,2]. The use of carbon-free fuels such as hydrogen on aircraft is limited by their substantial storage and handling requirements compared to hydrocarbon fuels, while the use of batteries significantly adds to the size and weight of the aircraft. Drop-in fuels derived from sustainable feedstocks are projected to be the most viable solution for abatement of aviation-related emissions in the coming decades [3]. Therefore, the adoption of scalable, sustainable, and economically viable fuels will be critical for mitigating the climate impact of the aviation sector. There is hence a need to predict the combustion behavior of novel, sustainable

aviation fuels (SAFs) at engine-relevant conditions.

Rapid and inexpensive assessment of fuel combustion performance over the entire operating regime of aircraft engines can be enabled by robust and compact chemical kinetic models. However, the molecular diversity and range of carbon numbers in conventional and alternative jet fuels make the development of accurate, detailed chemical kinetic models a challenging endeavor. These detailed models often contain thousands of reactions, which makes them computationally intractable. An alternate perspective towards the development of compact chemical kinetic models was conceived at Stanford, called the Hybrid Chemistry (HyChem) approach [4]. Conventional HyChem models rely on experimental constraints to accurately predict fuel combustion behavior under engine-relevant conditions. However, enabling the development of HyChem models for upcoming fuels without requiring experiments that demand large sample volumes could significantly accelerate SAF screening and deployment. This paper lays the foundation for a new

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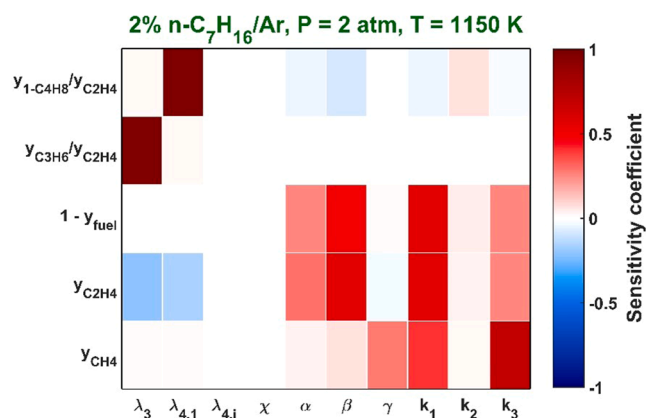


Fig. 1. Sensitivity of selected species' yields to the HyChem stoichiometric parameters and the rate coefficients of reactions (R1-R3) for 2 % *n*-heptane/Ar pyrolysis at 1150 K and 2 atm.

concept, namely IR-HyChem, which leverages the sensitivity of a fuel's combustion performance to its molecular structure to directly generate HyChem models based on the fuel's infrared (IR) spectrum.

As emphasized by Wang et al. [4], understanding the distribution of intermediates during fuel pyrolysis is critical for the development of HyChem models. To that end, our recent studies investigated the thermal decomposition of seven *n*- and isoparaffins behind reflected shock waves. Laser absorption spectroscopy-based diagnostics were employed to monitor the evolution of key pyrolysis products, i.e., methane, ethylene, and heavier C₃–C₄ alkenes [5–7]. The results demonstrated strong relationships in the high-temperature reactivity of hydrocarbons with their molecular structures. Here, a correlation has been established between the yields of these species and specific IR spectral features of the neat hydrocarbons. Using this correlation, HyChem models were developed for three real fuels (JP-8, RP-1, and F-24) based on their IR spectra. The performance of these models was subsequently evaluated against experimentally measured ignition delay times (IDTs). Moreover, a Monte Carlo analysis of the model uncertainties was performed, and possible strategies to further reduce uncertainty is discussed. Overall, the IR-HyChem approach is shown to hold promising potential to accelerate the screening and deployment of next-generation sustainable fuels.

2. Methodology

A detailed description of the Hybrid Chemistry (HyChem) modeling approach is provided elsewhere [4]. Briefly, the pyrolysis and oxidative pyrolysis part of a HyChem model can be defined for a typical hydrocarbon fuel by seven lumped reaction steps, with corresponding rate

coefficients (k_1 – k_7) and seven stoichiometric parameters (α , β , γ , λ_3 , $\lambda_{4,1}$, $\lambda_{4,i}$, χ).

HyChem models are traditionally constrained using speciation data obtained from flow reactor and shock tube experiments [4], owing to the strong sensitivity of the aforementioned parameters to the yields of stable intermediates during fuel pyrolysis. To illustrate this, Fig. 1 presents the sensitivity of the yields (at 1 ms) of key intermediates with respect to the seven independent stoichiometric parameters and the three reaction rate coefficients k_1 thru k_3 , corresponding to reactions (R1-R3), evaluated for 2 % *n*-heptane/Ar at 1150 K and 2 atm. Reactions (R4-R7) involve oxygenated species, and are therefore only pertinent to oxidative pyrolysis experiments. While such experiments serve to provide constraints for the rates k_4 thru k_7 , the current work only focuses on fuel pyrolysis without added O₂. Thus, only k_1 thru k_3 are considered here. All the intermediate yields are normalized by the initial fuel loading ($y_{\text{INT}} = x_i/x_{\text{fuel},0}$, where x_i is the mole fraction of a given intermediate, and $x_{\text{fuel},0}$ is the initial mole fraction of the fuel).

As seen in Fig. 1, the parameters α and β both exhibit a high sensitivity coefficient with respect to the ethylene yield ($y_{\text{C}_2\text{H}_4}$) and the fractional fuel consumption ($1 - y_{\text{fuel}}$). The rate coefficient k_1 , corresponding to the C–C fission of the fuel, understandably has notable sensitivity to the fuel consumption as well as the yields of key intermediates like C₂H₄ and CH₄. The rate of H-abstraction by CH₃ radicals (k_3), on the other hand, is dominantly sensitive to the methane yield. The parameter γ is, by definition, directly related to methane formation and is therefore solely sensitive to y_{CH_4} . As expected, λ_3 and $\lambda_{4,1}$ are strongly dependent on the ratio of propene and 1-butene yields to the C₂H₄ yield, respectively. It is evident from Fig. 1 that the HyChem stoichiometric parameters and reaction rates are sensitive to the yields of pyrolysis products. Hence, developing a comprehensive understanding of how various aspects of a fuel's molecular structure (e.g., carbon number, branching, and the presence of specific functional groups) impact the distribution of its pyrolysis products provides a viable alternative pathway for constraining HyChem models without the need for the pyrolysis speciation experiments. Unlike in the case of neat hydrocarbons, however, the presence of thousands of components in real fuels precludes the possibility of explicitly identifying all molecular structures.

Analytical techniques such as infrared spectroscopy can provide salient information about a complex fuel's composition and functional groups. To that end, this work utilizes the vapor-phase, Fourier Transform Infrared (FTIR) spectra of neat hydrocarbons and real fuels in the 2–15 μm wavelength range. Fig. 2 shows the representative FTIR spectra of nominal Jet-A fuel (A2) and three synthetic, alternative jet fuels (C1, C4, and C5). Compositional details and POSF numbers of these fuels are provided in the Supplementary Materials. As labeled in Figs. 2a, 2b, and 2c, different functional groups absorb infrared light at characteristic vibrational frequencies, and the absorption cross-section (CS) of a fuel at these frequencies is determined by the number of such functional groups

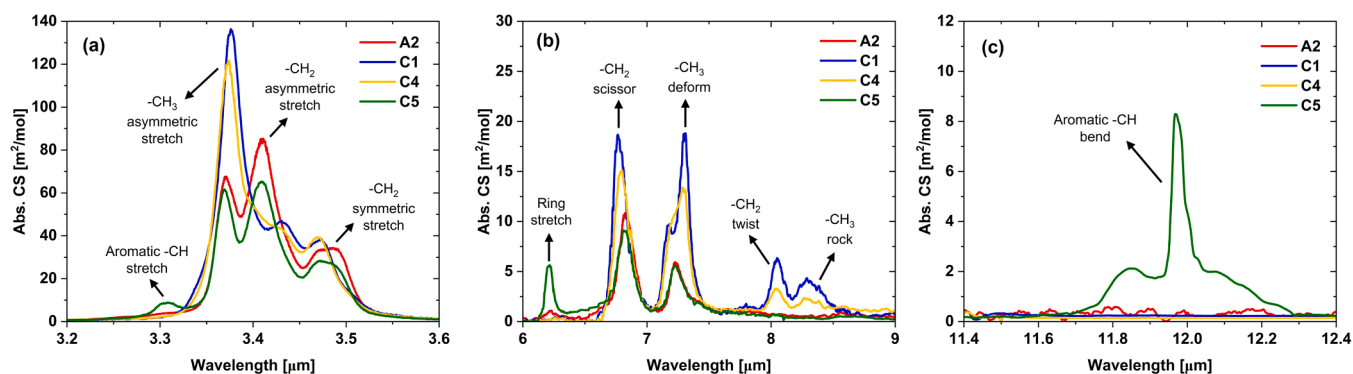


Fig. 2. Measured vapor-phase IR spectra of nominal Jet-A fuel (A2) and three synthetic, alternative jet fuels (C1, C4, and C5) revealing the important absorption features in three spectral regions: (a) 3.2–3.6 μm , (b) 6–9 μm , and (c) 11.4–12.4 μm .

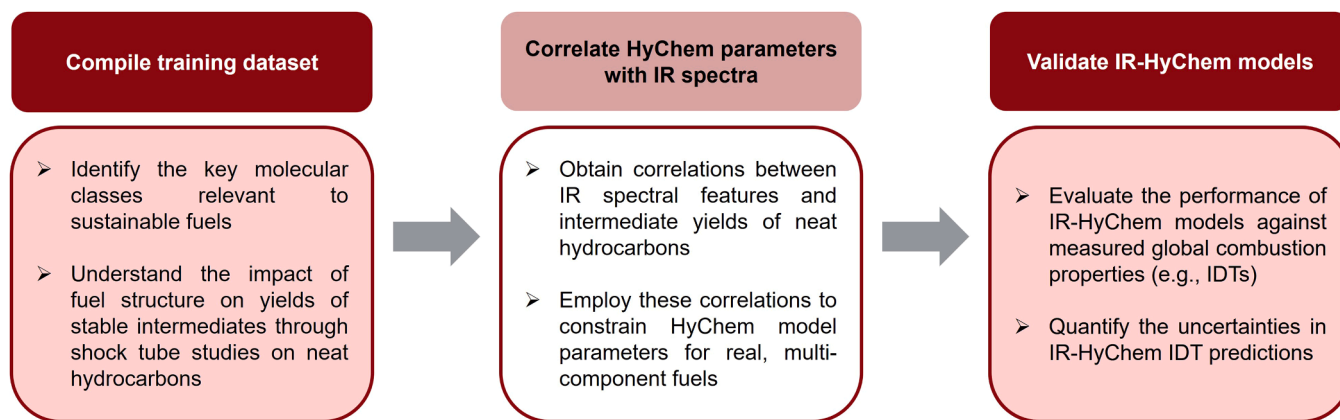


Fig. 3. Proposed methodology for developing and evaluating IR-HyChem models for real fuels.

present in the fuel. The IR spectrum can therefore serve as an effective proxy for the detailed composition of a fuel and can in principle be correlated with its pyrolysis product yields, and consequently with the HyChem model parameters.

Fig. 3 shows the proposed methodology for generating HyChem models for real fuels based on IR spectra. The first step involves compiling a training dataset of the FTIR spectra and pyrolysis product yields for a set of neat hydrocarbons spanning different molecular classes. Aviation fuels are typically comprised of four major molecular classes: *n*-paraffins, isoparaffins, cycloparaffins, and aromatics. Out of these, *n*- and isoparaffins are the key compounds of interest. In the current work, C_6 – C_8 hydrocarbons from these two classes were chosen to populate the preliminary training dataset. Ongoing work is focused on expanding the dataset to other molecular classes such as cycloparaffins, while also extending to higher carbon numbers (C_9 – C_{12}). Vapor-phase IR spectra of seven hydrocarbon fuels were measured over the 2–15 μm wavelength range using a Nicolet 6700 Fourier Transform Infrared (FTIR) spectrometer. Details about the shock tube pyrolysis experiments conducted to determine species yields are discussed in [5].

As shown in Fig. 3, the next step entails using the training dataset of neat hydrocarbons to obtain correlations between FTIR spectra and the measured intermediate yields. Since a given IR absorption feature is indicative of the same functional group in all fuels regardless of compositional complexity, these correlations could in principle be used also to predict intermediate yields for real, multi-component fuels, as long as the training dataset spans all the relevant molecular classes. The predicted intermediate yields can be used then to constrain HyChem parameters for a fuel under consideration.

In addition to constraining the parameters in the lumped reaction steps, building an IR-HyChem model also involves estimating the thermochemical properties of the fuel based on the IR spectra. For this, we utilized predictive regularized linear models. A similar approach to accurately predict the molecular weight, hydrogen-to-carbon ratio, and heat of combustion of fuels using IR spectra has been demonstrated in our previous work [6]. Finally, the constrained IR-HyChem models are evaluated against experimental measurements of global combustion properties, such as ignition delay times (IDTs) or laminar flame speeds. The current work employs IDTs measured at elevated pressures in a

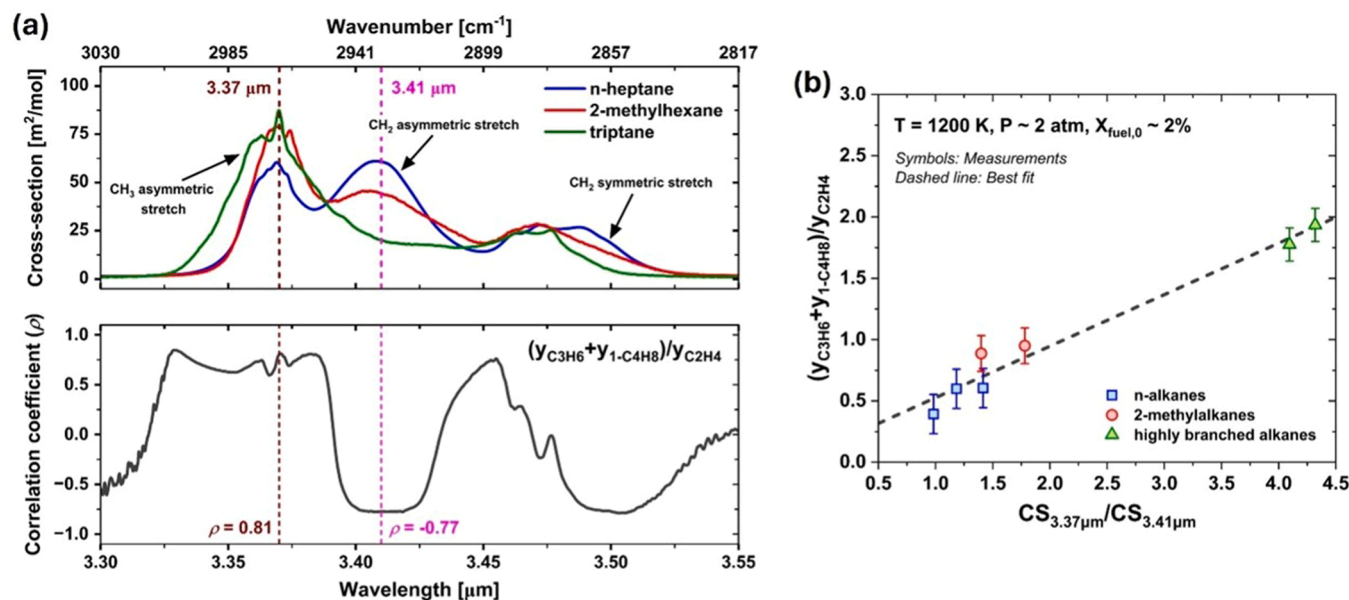


Fig. 4. (a) (Top) Representative infrared spectra of n-heptane, 2-methylhexane, and triptane in the 3.3–3.55 μm wavelength range. (Bottom) Pearson correlation coefficient calculated between the ratio of combined propene and 1-butene yields to ethylene yields (at $\sim 1200 \text{ K}$) and IR absorption cross-sections of the seven neat hydrocarbons considered in this work. (b) Ratio of combined propene and 1-butene yields to ethylene yields plotted against the ratio of absorption cross-sections at $3.37 \mu\text{m}$ (CH_3 asymmetric stretch) and $3.41 \mu\text{m}$ (CH_2 asymmetric stretch) for seven fuels studied at temperatures around 1200 K . Spectra of the seven neat hydrocarbons are shown in the Supplementary Materials.

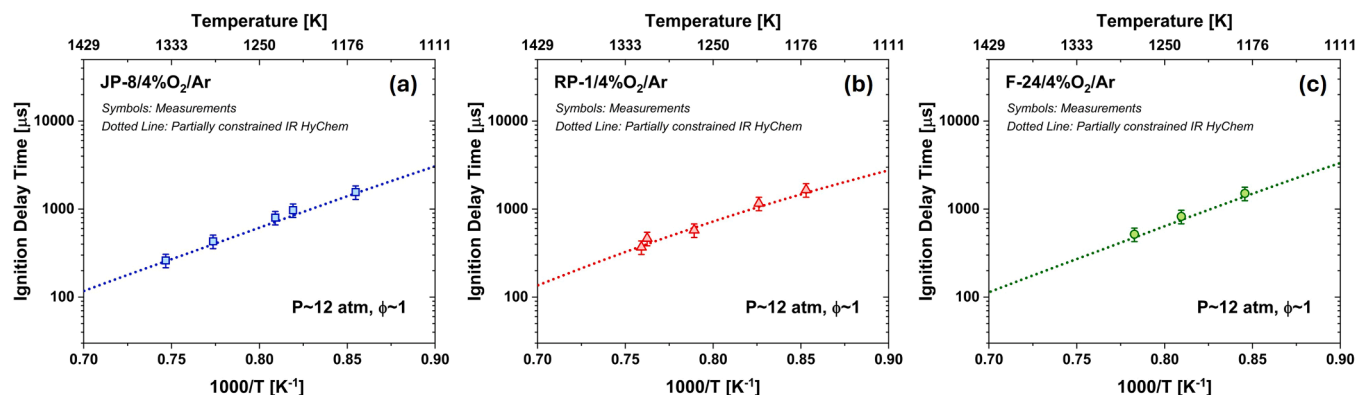


Fig. 5. Comparison of the measured ignition delay times (symbols) against partially constrained IR-HyChem model predictions (dotted lines) for (a) JP-8, (b) RP-1, and (c) F-24 in 4 % O_2/Ar , $\phi \sim 1$, $P \sim 12$ atm.

shock tube as validation targets.

Another important aspect of model validation is to quantify the uncertainty in the predicted IDTs. This can be achieved by performing Monte Carlo (MC) analysis, where each HyChem model parameter is randomly perturbed within its feasible range, subject to relevant constraints. The resulting variance in IDT predictions is indicative of the IR-HyChem model uncertainty. Such an analysis can also prove useful in revealing potential ways to improve model performance. In the following section, we follow this strategy to develop and evaluate IR-HyChem models for three real fuels (JP-8, RP-1, and F-24).

3. Results and discussion

3.1. Preliminary IR-spectra based correlations

Clear correlations in pyrolysis species yields with varying carbon numbers and degree of branching were observed during pyrolysis of neat hydrocarbons. These experiments were performed at a consistent pressure of 2 atm across a temperature range of 1100–1500 K. The effects of molecular structure on the correlations and stable intermediate formation have been thoroughly discussed in our recent studies [7,8]. In the present work, we correlate the ratio measured for the heavy alkene ($>C_2$) and ethylene yields with certain infrared spectral features to succinctly demonstrate the IR-HyChem methodology. The preliminary IR spectra-based correlation enables constraining the λ_3 and $\lambda_{4,1}$ parameters for the three real fuels. The resulting IR-HyChem models are partially constrained since not all species' yields were correlated with FTIR spectra. Developing correlations to constrain the remaining model parameters will be the focus of future work.

Fig. 4a (top panel) shows the IR spectra of *n*-heptane, 2-methylhexane, and triptane in the 3.3–3.55 μm “fingerprint” region, which contains spectral features corresponding to the C–H bond stretch in CH_2 and CH_3 groups. Key features include the asymmetric C–H stretch in CH_3 groups at 3.37 μm , the asymmetric C–H stretch in CH_2 groups at 3.41 μm , and the symmetric C–H stretch in CH_2 groups at 3.49 μm . As a straight-chain alkane, *n*-heptane exhibits the strongest absorption at wavelengths corresponding to CH_2 group features, while its absorption cross-section at 3.37 μm (corresponding to CH_3 groups) is the smallest among the three hydrocarbons. Conversely, triptane, which is a highly branched alkane, displays weaker CH_2 stretching features (3.41 μm and 3.49 μm) and a more pronounced CH_3 stretching feature.

The bottom panel of Fig. 4a illustrates the feasibility of obtaining pyrolysis product yields directly from IR spectra. Shown here is the Pearson correlation coefficient (ρ) calculated between the yield ratio of key intermediates, specifically, the combined yields of propene and 1-butene relative to ethylene, and the absorption cross-sections of seven fuels in the 3.3–3.55 μm region. The Pearson correlation coefficient quantifies the degree of linear relationship between two variables, with

values close to +1 or –1 indicating strong positive or negative correlations, respectively. Notably, spectral features near the CH_2 and CH_3 group absorption bands exhibit significant correlations, with $\rho = 0.81$ at 3.37 μm (CH_3 stretch) and $\rho = -0.77$ at 3.41 μm (CH_2 stretch). These findings suggest that the ratio of absorption cross-sections at these wavelengths can serve as a predictive metric for heavy alkene-to-ethylene yield ratios during pyrolysis. This is further supported in Fig. 4b, which shows a strong linear correlation between the combined 1 ms yields of propene and 1-butene, normalized by the ethylene yields ($Y_{C_3H_6} + Y_{1-C_4H_8} / Y_{C_2H_4}$) and the absorption cross-section ratio at 3.37 μm and 3.41 μm ($CS_{3.37\mu m} / CS_{3.41\mu m}$) for seven neat hydrocarbons at 1200 K. The dashed line in Fig. 4b indicates the best fit obtained between these two variables. This offers the opportunity to predict the ratio of combined propene and 1-butene yields to ethylene yields for a real fuel simply based on the ratio of its absorption cross-sections at 3.37 μm and 3.41 μm . Since $(Y_{C_3H_6} + Y_{1-C_4H_8} / Y_{C_2H_4})$ relates to the sum of λ_3 and $\lambda_{4,1}$, the linear correlation shown can be used to constrain $\lambda_3 + \lambda_{4,1}$ for real fuels. After measuring the FTIR spectra of the three real fuels, the linear fit resulted in the following constraints on $\lambda_3 + \lambda_{4,1}$: (a) JP-8: 0.49, (b) RP-1: 0.45, and (c) F-24: 0.44. The remaining HyChem parameters were obtained using previously developed models for jet and rocket fuels whose molecular compositions were similar to the three fuels, details of which can be found in Sections 3, 5 and 6 of the [Supplementary Material](#).

3.2. Model performance and uncertainty analysis

Ignition delay times (IDTs) measured around 12 atm over the temperature range of 1150–1400 K were used as validation targets for the partially constrained IR-HyChem models developed in the current work. These experiments were conducted in Stanford's High-Pressure Shock Tube (HPST) facility using manometrically prepared, stoichiometric mixtures of the three real fuels in 4 % O_2/Ar . The IDTs were quantified using an OH^* emission diagnostic, as described in [9]. Fig. 5 shows the IDTs measured for (a) JP-8, (b) RP-1, and (c) F-24 compared against the predictions of their respective IR-HyChem models. It is evident that the partially constrained IR-HyChem models accurately predict the IDTs of all three fuels considered over the entire temperature range. It should be noted that although the intermediate yields used to develop the correlations were measured around 2 atm, the IR-HyChem models are found to perform well at much higher pressures (~ 12 atm), illustrating the applicability of these models over a wide spectrum of engine-relevant operating conditions.

The accuracy of an IR-HyChem model depends on the degree to which the various model parameters are constrained [10]. Each independent stoichiometric parameter in the HyChem model can take values within a feasible range subject to elemental balance. Incorporating additional constraints leads to tighter bounds on these valid parameter

Table 1

Nominal values and parameter ranges used in MC sampling analysis for the two test cases.

Parameter	Nominal values	MC parameter range	
		Case 1: Unconstrained	Case 2: Partially constrained
α	0.5	[0,2]	[0,2]
β	0.3	[0,1]	[0,1]
γ	0.45	[0, $\gamma_{\max}^*$]	[0, $\gamma_{\max}^*$]
λ_3	0.37	[0,2]	[0.33, 0.41]
$\lambda_{4,1}$	0.12	[0,2]	[0.10, 0.14]
$\lambda_{4,i}$	0.05	[0,2]	[0,2]
χ	0.51	[0,1]	[0,1]

$$*\gamma_{\max} = \frac{[(\chi - 4)m + \frac{1}{2}(7 - \chi) + 3\beta]}{10 - \chi} - 1, m = \text{Carbon number.}$$

sets. Since the stoichiometric parameters are obtained through IR spectra-based correlations, they are subject to measurement uncertainties in both the pyrolysis product yields and the FTIR absorption cross-sections. This means that the constrained parameters are not unique, and there exists a feasible set within which all combinations of parameter values can reproduce the species' yields used to constrain the model in the first place.

To quantify the uncertainty in the predicted global combustion

properties due to the uncertainties in model parameters, we carried out Monte Carlo (MC) analysis by generating 500 randomly sampled IR-HyChem models. JP-8 was chosen as the test fuel for this analysis. In each model, the parameters were randomly selected within their respective feasible sets for two cases: (1) unconstrained and (2) partially constrained. Both models use the same nominal values of all the stoichiometric and kinetic parameters. However, the unconstrained case is essentially only subject to element balance (see [10] for a detailed discussion). The partially constrained model, on the other hand, has its λ_3 and $\lambda_{4,1}$ constrained using the correlation described in the preceding subsection. Therefore, these two parameters now have tighter bounds on their feasible sets, determined based on the measurement uncertainty in the intermediate yields as well as the FTIR spectra. Our analysis revealed that the net uncertainty on the combined prediction of $(y_{C_3H_6} + y_{1-C_4H_8})/y_{C_2H_4}$ is 12.5 %. Using this, $\lambda_3 + \lambda_{4,1}$ was allowed to vary within [0.43, 0.55] for JP-8. Observing the variance in the IDT predictions of both these models elucidates the value of imposing constraints on different model parameters.

Table 1 lists the nominal values of the seven stoichiometric parameters for the JP-8 IR-HyChem model, accompanied by the feasible parameter sets for the unconstrained and partially constrained cases. In the absence of any constraints, the ratio parameters (λ_3 , $\lambda_{4,1}$, and $\lambda_{4,i}$) are sampled from the ranges that correspond to random productions of C_2H_4 , C_3H_6 , $1-C_4H_8$, and $i-C_4H_8$ such that the mean ratio values equal unity. Furthermore, the pre-exponential factors of rates k_1 thru k_7 were

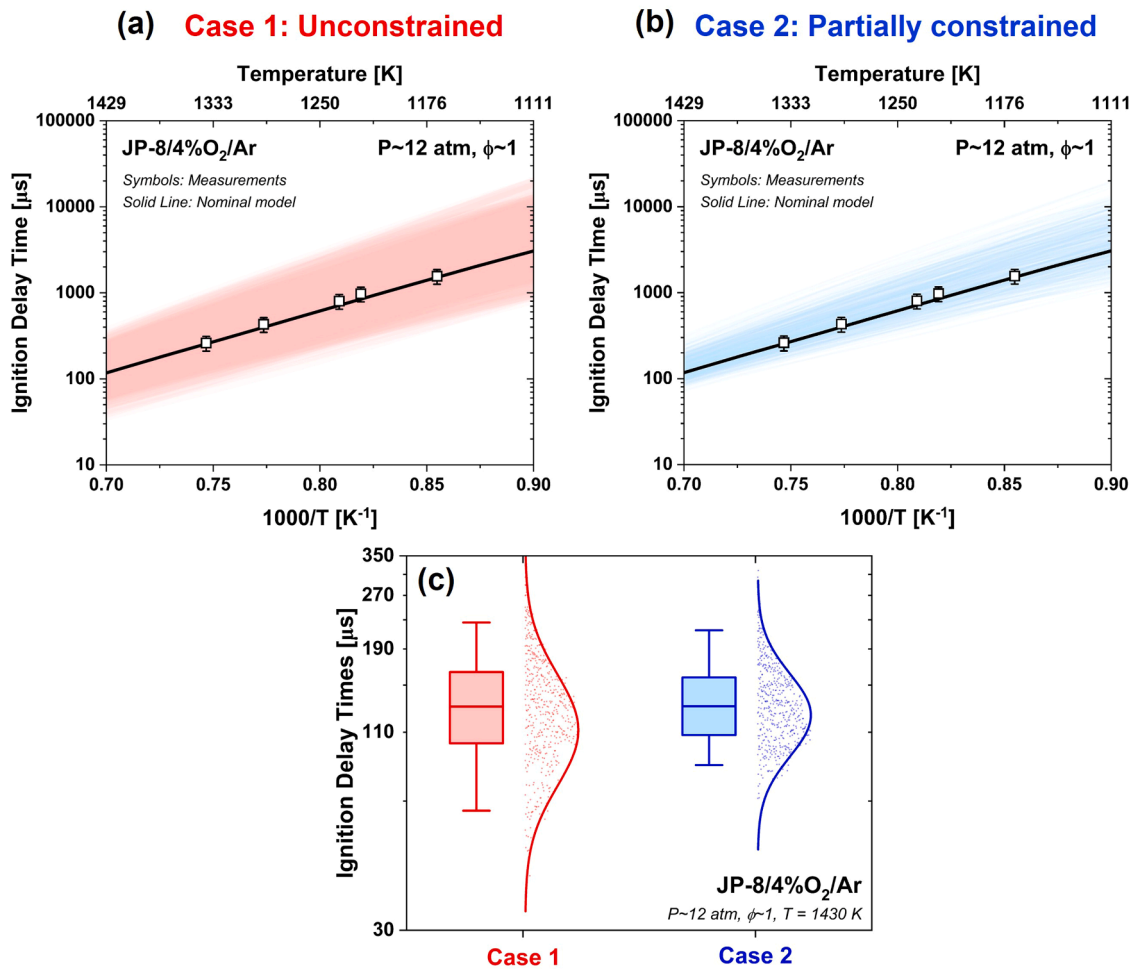


Fig. 6. Uncertainty in predicted IDTs (JP-8/4 % O₂/Ar, $\phi = 1$, $P = 12$ atm) for (a) unconstrained and (b) partially constrained IR-HyChem models. The faded bands around the nominal model predictions represent the variance in IDT predictions observed across the 500 randomly sampled MC cases. (c) Box-and-whisker plot of the MC simulations showing that spectra-based constraints reduce prediction uncertainty, with whiskers representing 95 % confidence intervals. Boxes span the interquartile range (25–75 percentile) and indicate the median line.

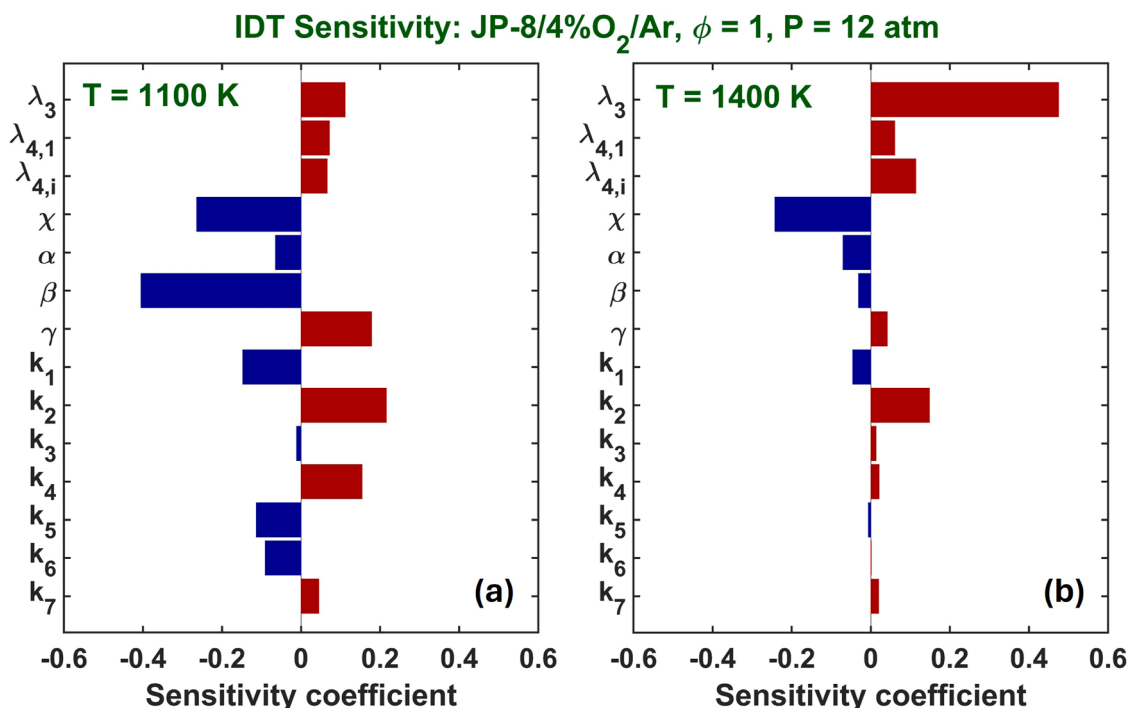


Fig. 7. Sensitivity of IDTs (JP-8/4 % O₂/Ar, $\phi = 1$, $P = 12$ atm) to the HyChem stoichiometric parameters and rate coefficients at (a) 1100 K and (b) 1400 K.

perturbed within a factor of five from their respective nominal values, given the lack of any data-driven kinetic constraints.

Figs. 6a and 6b show the results for the unconstrained and partially constrained MC cases, respectively. Although the uncertainty in model predictions is found to be larger than experimental uncertainty (± 15 %) for both the MC cases, the partially constrained model still shows improved performance compared to the unconstrained model. In particular, the 95 % confidence interval of the partially constrained model predictions is smaller by ~ 5 % at 1120 K, and almost 30 % at 1420 K compared to the unconstrained case (refer Fig. 6c). This underscores the impact that constraining even a subset of IR-HyChem model parameters can have on the overall uncertainty in combustion property predictions.

To identify potential ways to reduce model uncertainties, we conducted a brute-force sensitivity analysis of IDTs with respect to each stoichiometric parameter and rate coefficient. The sensitivity coefficients were evaluated at two temperatures: 1100 K and 1400 K, as shown in Figs. 7a and 7b, respectively. As it can be seen, at 1400 K, IDT is dominantly sensitive to the parameter λ_3 . Since the partially constrained model imposes tighter bounds on λ_3 , the reduction in uncertainty is understandably greatest at higher temperatures. At 1100 K, however, the sensitivity to λ_3 is diminished, with the parameters β , χ , γ , and the rate coefficients k_1 , k_2 , and k_4 showing enhanced sensitivity. Since no constraints have been imposed on these parameters in the present work, the reduction in model uncertainty is not as pronounced at lower temperatures. This analysis clearly suggests the potential of further minimizing the uncertainty in IR-HyChem model predictions by acquiring additional infrared spectra-based correlations to constrain the remaining stoichiometric and kinetic parameters. For instance, the parameter β , which shows the highest sensitivity at the lower temperature case can be constrained using absolute ethylene yields (refer to Fig. 1) predicted using the infrared spectra. This will be a key objective of future work to refine and extend the IR-HyChem methodology.

4. Summary and conclusions

To enable low-volume characterization of next-generation

sustainable fuels, a novel methodology, termed IR-HyChem, was proposed for modeling the combustion behavior of real fuels based on their infrared (IR) spectra. A training dataset of seven neat hydrocarbons, including normal and branched alkanes, was compiled using their vapor-phase, 2–15 μm Fourier Transform Infrared (FTIR) spectra. Recently, shock-tube experiments were conducted to study the time-resolved evolution of key stable intermediates: methane, ethylene, and heavier alkenes ($>C_2$) during the pyrolysis of these neat hydrocarbons utilizing laser absorption spectroscopy (LAS). These measurements allowed the inference of clear correlations between the yields of stable intermediates and the molecular structure of the fuel, providing valuable insights into fuel reactivity under combustor-relevant conditions. A strong linear correlation was obtained between the ratio of heavy alkene ($>C_2$) to ethylene yields and the ratio of IR absorption cross-sections at two wavelengths corresponding to CH₃ and CH₂ functional groups. This correlation was employed to predict the sum of stoichiometric parameters λ_3 and $\lambda_{4,1}$ for three real fuels: JP-8, RP-1, and F-24, resulting in partially constrained IR-HyChem models for these fuels. The ignition delay times (IDTs) predicted by these models were found to agree well with experimental measurements for all three fuels. A Monte Carlo uncertainty analysis was performed by randomly sampling HyChem parameters within their respective feasible sets for two test cases: (1) unconstrained and (2) partially constrained IR-HyChem models of JP-8. Since the unconstrained model was only subject to element balance, it showed higher variance in predicted IDTs compared to the partially constrained case, which imposed constraints on λ_3 and $\lambda_{4,1}$ derived from FTIR spectra. A brute-force sensitivity analysis for IDTs suggested that the model uncertainties could be further minimized by developing additional IR spectra-based correlations to constrain the remaining stoichiometric parameters and rate coefficients. Overall, the IR-HyChem approach holds the potential to reduce the test fuel volumes needed and costs associated with the regulatory testing and approval process of sustainable fuel candidates.

Novelty and significance statement

This work addresses the urgent need for rapid, low-volume

assessment of sustainable aviation fuels (SAFs). A critical step toward this goal is the accurate combustion modeling of emerging SAFs, which is enabled by compact kinetic models such as the HyChem approach. The study introduces a novel framework, namely IR-HyChem, which for the first time leverages a real fuel's vapor-phase FTIR spectrum to directly inform the development of their HyChem models. Unlike traditional methods that require extensive experimental data and large fuel samples, this method uses empirical correlations between FTIR spectral features and the yields of key pyrolysis intermediates, measured through sensitive laser-based diagnostics, to constrain model parameters. This approach is demonstrated on three real fuels and shows excellent performance against elevated-pressure ignition delay time measurements. The authors believe that this integration of spectroscopy, kinetics, and data-driven modeling establishes a new, low-volume (<1 mL) methodology in SAF evaluation under engine-relevant conditions.

CRediT authorship contribution statement

Pujan Biswas: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Vivek Boddapati:** Writing – review & editing, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Andrew R. Klingberg:** Investigation, Data curation. **Alka Panda:** Investigation, Data curation. **Hai Wang:** Writing – review & editing, Supervision, Funding acquisition. **Ronald K. Hanson:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.proci.2025.105792](https://doi.org/10.1016/j.proci.2025.105792).

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